Predicting Coronary Heart Disease from Survey Data with Linear Models
Machine Learning CS-433, Project 1

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I. Introduction

We implement six linear methods from scratch in NumPy and use them to predict coronary heart disease from the BRFSS 2015 telephone health survey which has 328,135 training samples and 321 features. Only 8.8% of the respondents are positive, so we decided to optimise the F1 score rather than accuracy. Our best submission, regularised logistic regression with survey-specific preprocessing and a tuned decision threshold, reaches an F1 score of 0.452 on AIcrowd.

II. Implemented functions

We implemented the six required methods: linear regression with gradient descent `mean_squared_error_gd` and stochastic gradient descent with batch size 1 in `mean_squared_error_sgd`, least squares and ridge regression via the normal equations, and logistic regression without and with an L_2 penalty $\lambda\|w\|^2$, both trained with gradient descent. All functions return the final weights and the loss without the penalty term. They reproduce the expected values of the official grading tests, and pass the 9 additional unit tests we wrote in `test_implementations.py`.

III. Data Engineering

Not all features in the provided dataset were ready to be used by our models and had to be preprocessed.

A. Dataset specific preprocessing

We started by deleting features classified as metadata, as they do not provide information helpful for classification. In total we removed 19 such columns, such as dates, sequence numbers and survey weights, listed in `feature_engineering.py`.

Subsequently, we decoded the survey's special answer codes. Their meaning depends on the number of digits of a column's answers, which we read from the column's maximum value in the training set like 7/77/777 ("don't know") and 9/99/999 ("refused") were replaced by `NaN`, and 88/888 ("none", e.g. zero days) by 0. Without this step, the number of days in poor physical health has a mean of 60.7 days per month instead of 4.3.

B. General preprocessing

Columns with 3 to 10 distinct values are categorical answers, e.g. general health 1-5, so we one-hot encoded them, leaving out the first value to avoid a column that duplicates the bias. In the remaining numerical or binary columns we replaced `NaN` by the median. Since missing answers are not random but caused by the survey's skip logic e.g. the question about blood pressure medication is only asked to people with high blood pressure, we added for every column with missing values an indicator column `x_missing`. Finally, we removed constant columns, standardised every feature and added a bias column, which gives 857 features. All statistics, medians, means, and standard deviations were computed on the training set only and reused for the test set.

IV. Results

We selected models on a single stratified 80/20 split of the training data, so that both parts keep the 8.8% positive rate. All preprocessing statistics were computed on the 80% part. For the final AIcrowd submissions, we retrained the models on all labelled training samples, recomputing the preprocessing stats on the full training set. For ridge regression we tried $\lambda \in \{10^{-5}, 10^{-4}, 10^{-3}\}$; for regularised logistic regression the same values of λ with $\gamma = 0.5$ and 1000 gradient descent iterations. For every model we chose the decision threshold on $x^\top w$ that maximises the validation F1, searching from -2 to 1 in steps of 0.05. Table I compares the methods and Fig. 1 shows how strongly F1 depends on the threshold. With the default threshold almost all samples are predicted negative because of the class imbalance.

Our preprocessing improves the AIcrowd F1 of regularised logistic regression from 0.412 with median imputation and standardisation only to 0.451, a gain of 0.039, which is larger than the gain of switching from ridge regression 0.439 to logistic regression 0.012. The configuration with the highest validation F1 0.440 scored 0.451 on AIcrowd. Our best AIcrowd score, 0.452, came from a manually chosen configuration ($\gamma = 0.5$, $\lambda = 10^{-4}$, 1000 iterations, threshold -1.35 on $x^\top w$, i.e. a positive prediction when $\sigma(x^\top w) > 0.21$), which ranked third on validation F1 0.439. Since this choice was made on the public leaderboard, its

Table I
VALIDATION AND AICROWD F1 (BEST THRESHOLD PER MODEL).

Model	Val. F1	AIcrowd F1
Reg. log. reg., baseline preprocessing*	–	0.412
Ridge regression + our preprocessing	0.429	0.439
Reg. log. reg. + our preprocessing (best val.)	0.440	0.451
Reg. log. reg. + our preprocessing (manual)	0.439	**0.452**

*median imputation and standardisation only

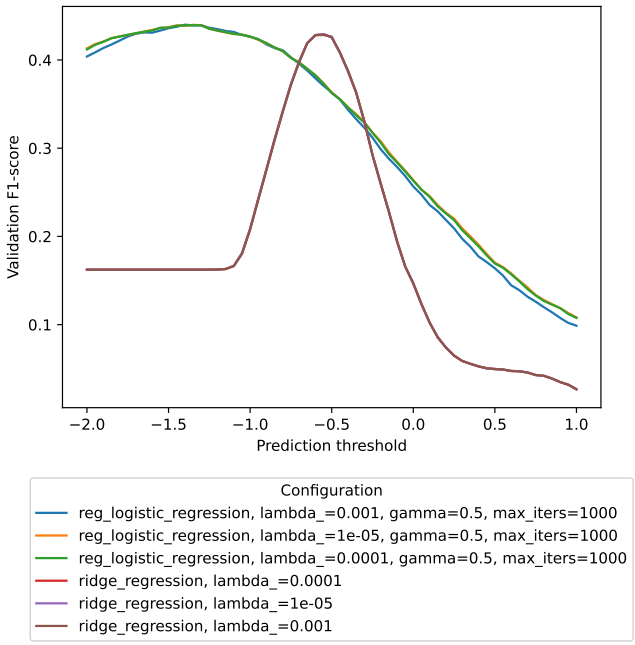


Figure 1. Validation F1 as a function of the decision threshold on $x^\top w$ for several logistic and ridge regression configurations. For ridge regression the scores approximate the labels ± 1, for logistic regression they are log-odds.

score is optimistically biased; the difference of 0.001 to the validation-selected model is too small to prefer either.

V. CONCLUSION

Decoding the survey's answer codes and treating missing answers as information mattered more than the choice of the linear model, and tuning the decision threshold was essential because of the class imbalance. With more time, we would replace the single validation split by k-fold cross-validation to obtain more reliable model selection, and add polynomial features of the continuous variables such as age.